

LYMPHOID NEOPLASIA

Inhibition of the atypical kinase WNK1 as a therapeutic strategy in TAL-related T-cell acute lymphoblastic leukemia

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KEY POINTS

- **WNK1 is identified as a promising therapeutic target in T-ALL and its inhibition induces mitotic defects.**
- **WNK1 is overexpressed in the TAL1-related T-ALL subgroup, suggesting a selective therapeutic window.**

Driver mutations in T-cell acute lymphoblastic leukemia (T-ALL) rarely affect druggable kinases. However, these kinases can be aberrantly activated or repressed as secondary oncogenic events. Thus, integrating unbiased phosphoproteomics with genomic approaches may offer novel opportunities for target discovery and therapeutic interventions. In our study, we identified WNK1 (with no lysine [K]) as a potential target in T-ALL by pairing a list of vulnerable kinases with data from a phosphoproteomic screen of T-ALL cell lines. We subsequently validated WNK1 by loss-of-function-based studies and tested WNK inhibitors in several in vitro and in vivo T-ALL models and clinical T-ALL samples. We showed that therapeutic WNK1 repression promotes polyploidy, resulting in cell proliferation arrest, and morphometric changes, such as incomplete cell division or chromosome segregation through altered mitotic spindle assembly and abscission defects. Furthermore, we found that WNK1 is overexpressed in the TAL1/2-related subgroup, but not in normal thymus or lymph nodes, suggesting a potential translational area for clinical exploitation in poor-prognosis T-ALL carrying *PTEN* mutations and *del(6q)*. Our work also reports a functional contribution of WNK1 in the leukemia establishment and progression. Structurally WNK1 is an atypical serine/threonine kinase that diverges from canonical kinases by lacking the conserved lysine in subdomain II, instead featuring a cysteine in subdomain I, which is critical for adenosine triphosphate (ATP) binding. This unusual structural configuration creates a distinct ATP-binding pocket with limited sequence similarity to conventional kinases, offering a unique opportunity to develop highly selective small molecules. Targeting this atypical ATP domain could thus provide a therapeutic advantage and broaden the treatment landscape for T-ALL.

Introduction

The identification of new cancer drug targets has been driven in part by the advent of next-generation sequencing technologies that have revealed aberrant mutations, which can be pharmacologically targeted due to their activating nature. This approach has revolutionized some therapeutic areas, such as the treatment of chronic myeloid leukemia, where an aberrant kinase fusion corresponds directly to a targeted drug.¹⁻³ The complexity of cancer initiation and progression, however, has tempered the easy matching of activating mutation with

molecular therapies, especially in cancers where the mutational load in genes coding for easily targetable proteins is low. This is, for example, the case with malignancies such as T-cell acute lymphoblastic leukemia (T-ALL), which are dependent on the activation of aberrantly overexpressed lineage-dependent genes (ie, transcription factors) or on poorly druggable mutants.^{4,5} Given the paucity of targetable kinases across most T-ALL subgroups and their poor outcomes, especially in relapsed/refractory cases, T-ALL remains a strong model for advancing multiomic screening-based strategies for target identification.

In this study, to characterize druggable kinases active regardless of their mutational status, we integrated the largest proteomic T-ALL data set⁶ with the analysis of a T-ALL CRISPR dependency screen⁷ and pinpointed the role of the kinase WNK1. WNK (with no lysine [K]) proteins (WNK1-4) are atypical serine/threonine kinases in which the catalytic lysine, necessary for adenosine triphosphate (ATP) binding, is replaced by cysteine and located in subdomain I instead of II.⁸ In mammals, WNK kinases regulate ion homeostasis and blood pressure regulation⁸⁻¹⁰ by modulating the kinases SPAK (STE20/SPS1-related proline-alanine-rich protein kinase) and OSR1 (oxidative stress-responsive kinase).⁹ Gain-of-function mutations in *WNK1* and *WNK4* cause pseudohypaldosteronism type II, a disorder marked by hyperkalemia and hypertension.¹¹

In solid tumors, WNK proteins have been mainly implicated as potential clinical biomarkers of aggressiveness or tumor invasiveness.^{12,13} The role of WNK1 in hematological malignancies is largely unknown, except for multiple myeloma where WNK1 inhibition has been reported to reduce *MYC* transcription and disrupts the activity of the *IgH* enhancer, or in acute myeloid leukemia (AML) where it controls the mammalian target of rapamycin complex 1 (mTORC1) signaling by regulating amino acid uptake through phosphorylation of amino acid transporters, particularly SLC38A2.^{14,15}

In this study, we show that WNK1 appears to preferentially sustain the growth of the largest T-ALL subtype, *TAL1/2*. Given the uniqueness of the WNK kinase pocket, it represents as a potential alternative nonredundant therapeutic target in this disease.

Methods

Primary T-ALL samples and T-lymphocytes

Peripheral blood (PB) and bone marrow (BM) leukemic cells or lymphocytes were obtained under an approved protocol at the Department of Medicine and Surgery at Parma University Hospital (no. 265/2019 and no. 767/2022) from individuals affected by T-ALL or healthy blood donors, respectively. All procedures were performed according to the Declaration of Helsinki guidelines. All biological samples were collected after the release of a written informed consent.

T-ALL in vivo studies

Patients-derived leukemia xenografts (PDLXs) were established by tail vein injection of mononucleated cells isolated from T-ALL primary samples into male NOD.Cg-Prkdcscid IL2rgtm1Wjl/SzJ (NSG) mice aged 6 weeks. Procedures were approved under the protocols N.682/2019-PR, 137/2024-PR, and 176/2025-PR.

For the WNK463 efficacy experiment, mice were randomized for treatment starting 5 days after transplant. WNK463 was dissolved in 10% dimethyl sulfoxide and 90% corn oil and administered via oral gavage at 1.5 mg/kg per day dose for 21 days.

STIL::*TAL1* cells were transduced with a tetracycline-inducible short hairpin RNA (shRNA) (#45 or nontargeting [NT])

targeting *WNK1* and injected into mice. Doxycycline (2 mg/mL) was administered in drinking water to induce shRNA expression. A control group (shRNA #45) did not receive doxycycline.

All other methods are described in detail in the supplemental Methods, available on the *Blood* website.

Results

To profile kinase dependencies in T-ALL, we intersected the most extensive proteomic analysis of 11 T-ALL cell lines⁶ with the genomic and functional data sets available at the Cancer DepMap Project (DepMap).⁷ Of the 213 highly expressed proteins in T-ALL⁶ with a gene dependency score of less than -1, WNK1 and CDK1 shared the highest mRNA and protein expression abundance value (Figure 1A; supplemental Table 1.1-1.4). We then restricted our analysis to 91 kinases, which were reported to be “drivers” in cancer,¹⁶ and identified a strong *WNK1* dependency in T-ALLs (Figure 1B; supplemental Table 1.5-1.6). We used the FORALL data sets, which integrate proteomic, transcriptomic, and pharmacological profiling of 22 T-ALL cell lines,¹⁷ to minimize discrepancies between mRNA and protein levels and reduce the risk of overfitting for transcriptional-based gene dependencies. As shown in supplemental Figure 1A, WNK1 protein levels and mRNA expression highly correlate (Pearson $r = 0.64$, $P = .00123$, $n = 22$), suggesting that WNK1-dependent effects can be reliably captured through mRNA expression analysis.

WNK1 is a member of a family of 4 serine/threonine protein kinases.⁸ To investigate whether *WNK1-4* are expressed at similar levels in T-ALL, we analyzed transcriptional data from 287 pediatric patients¹⁸ (Figure 1C), as well as the expression in T-ALL cell lines, primary T-ALL samples,¹⁹ and PDLX T-ALL cells (Figure 1D). Our findings reveal that *WNK1* is differentially expressed compared with other WNK family members (*WNK1* > *WNK2* > *WNK4* > *WNK3*; Figure 1C) and is not correlated, for example, with *WNK2* (supplemental Figure 1B-D). Cross-validation with the RNAi gene enrichment ranking²⁰ confirmed that *WNK1*, and not *WNK2*, suppression preferentially induces T-ALL cell death, supporting its targeting in this disease (Figure 1E; supplemental Figure 1E-F).

We also demonstrated significantly higher *WNK1* transcript levels in primary T-ALL samples compared with CD3⁺ T-lymphocytes (Figure 1F-G; supplemental Figure 1G; supplemental Table 2) and compared with B-cell acute lymphoblastic leukemia and AML cell lines (Figure 1H), suggesting the possibility of a higher therapeutic index for T-ALL compared with other acute leukemia subtypes.

In conclusion, at the nexus of functional screens, WNK1 emerged as a key therapeutic target in T-ALL.

WNK1 expression characterizes the *TAL1/2* subgroup in T-ALL

Given the heterogeneous expression of WNK1 in T-ALL (Figure 1F; supplemental Figure 2A), we next asked whether variation in WNK1 levels defines a genetic subset of T-ALL in which its expression is preferentially higher than in other subtypes. From the analysis of pediatric and adult T-ALL data

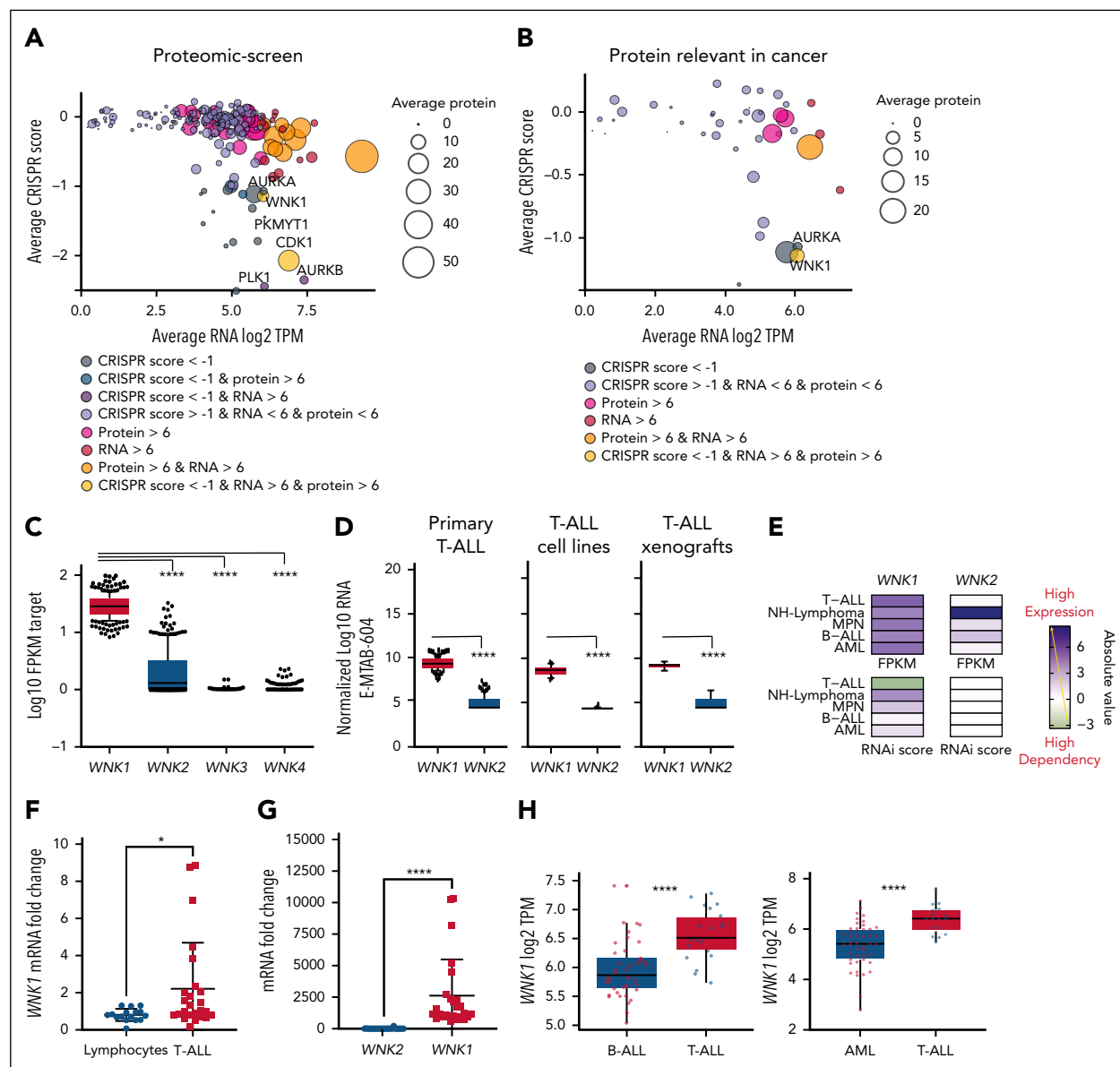


Figure 1. Identification of WNK1 in T-ALL. (A) The scatterplot displays the relationship between RNA expression levels (x-axis, log₂ transcripts per million [TPM]) of 213 protein kinases in T-ALL cell lines (n = 11) and their CRISPR screen (DepMap Public 24Q2) dependency scores (y-axis, average CRISPR score). Each point represents an individual gene, with the size of the dots corresponding to the average protein levels, as shown by the bubble scale on the right. Larger circles represent proteins with higher average expression levels. High-confidence targets displaying a CRISPR dependency score of less than -1, mRNA log₂ TPM expression >6, and protein expression >6 normalized protein lysate counts were depicted in yellow. (B) The scatterplot displays the relationship between RNA expression levels (x-axis, log₂ TPM) of a list of 91 protein kinases putative “driver” in cancer¹⁶ in T-ALL cell lines and their CRISPR screen (DepMap Public 24Q2) dependency scores (y-axis, average CRISPR score). Each point represents an individual gene, with the size of the dots corresponding to the average protein levels, as shown by the bubble scale on the right. Larger circles represent proteins with higher average expression levels. High-confidence targets displaying a CRISPR dependency score of less than -1, mRNA log₂ TPM expression >6, and protein expression >6 normalized protein lysate counts were depicted in yellow. (C) Box-and-whisker plot showing WNKs (WNK1-4) log₁₀ fragments per kilobase (FPKM) mRNA expression in 287 pediatric T-ALL samples. The box represents the interquartile range containing the middle 50% of the data; the whisker, the lines extending from the box, represents the range of the data, excluding outliers, which are marked individually. Statistical significance among groups (****P ≤ .0001) was determined by a 1-way analysis of variance (ANOVA) using Dunnett’s correction for multiple comparison testing. (D) Box-and-whisker plot displaying WNK1 (red) and WNK2 (blue) expression in primary T-ALL (n = 73), T-ALL cell lines (n = 20), and T-ALL xenografts (n = 9; E-MTAB-604). Statistical significance among groups (****P ≤ .0001) was determined by Mann-Whitney U test. (E) Heat maps showing gene expression (FPKM, top panels) and RNAi dependency scores (bottom panels) for WNK1 and WNK2 across hematological malignancies: T-ALL, non-Hodgkin lymphoma, myeloproliferative neoplasms, B-cell acute lymphoblastic leukemia (B-ALL), and AML. Color intensity represents absolute values, as indicated by the scale bar on the right. (F) Scatterplot showing WNK1 mRNA relative to the control gene RPL13A in lymphocytes (n = 16; blue) and primary T-ALL samples (n = 25; red). Statistical significance among groups (*P ≤ .05) was determined by Mann-Whitney U test. (G) Scatterplot showing WNK1 (red) and WNK2 (blue) mRNA relative to the control gene RPL13A in primary T-ALL samples (n = 25). Statistical significance among groups (****P ≤ .0001) was determined by Mann-Whitney U test. (H) Box-and-whisker plot displaying WNK1 mRNA expression in T-ALL (red) and B-ALL (blue) cell lines (left) and AML (right). Data were obtained from the following data sets: FORALL¹⁷ and DepMAP⁷ projects. Statistical significance among groups (****P ≤ .0001) was determined by Mann-Whitney U test.

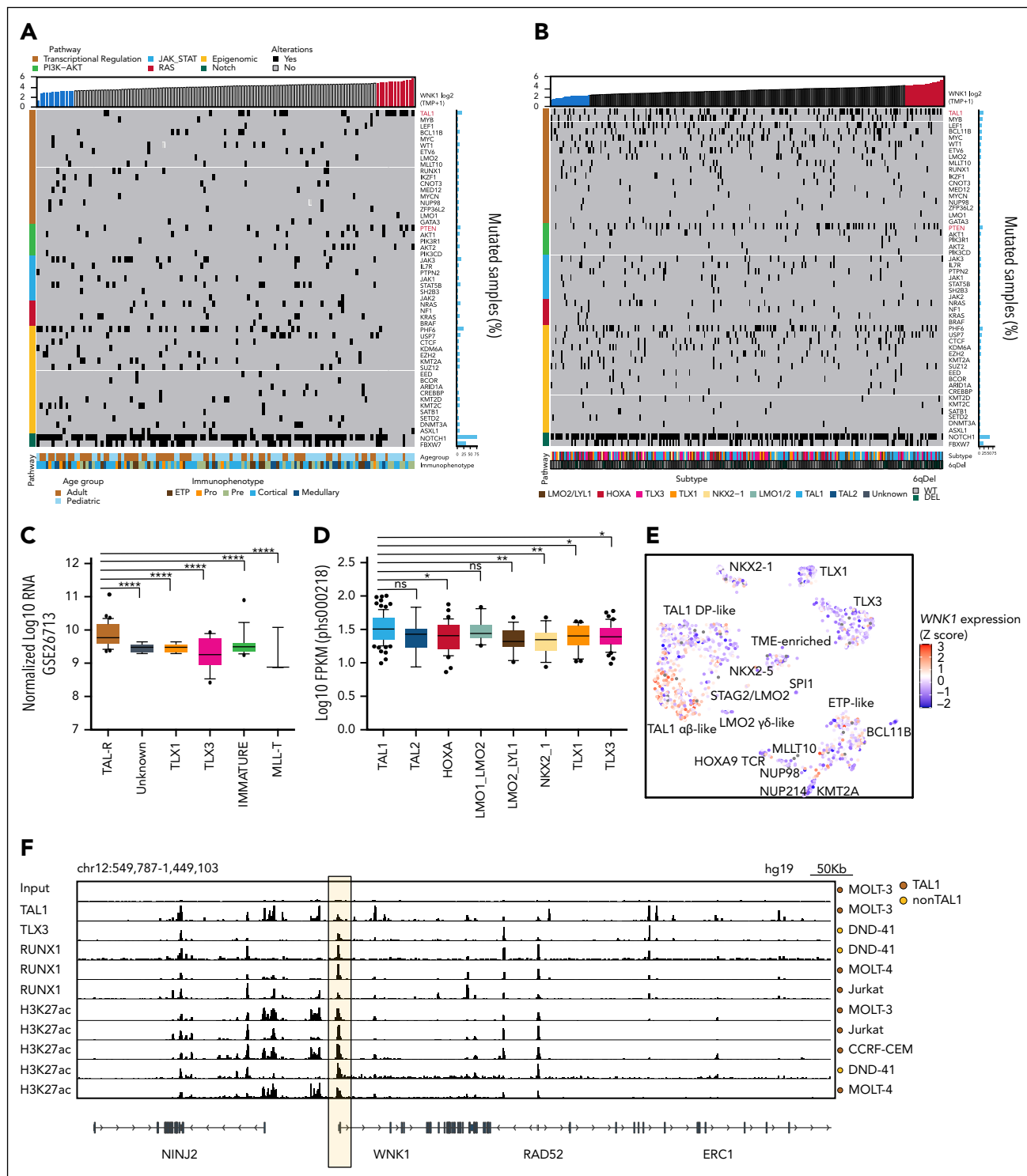


Figure 2. *WNK1* is overexpressed in the TAL-R subgroup. (A) The heat map shows the OncoPrint distribution of genetic alterations across key signaling pathways in 61 adult and 69 pediatric T-ALL samples.²¹ Rows represent individual genes, and columns correspond to patient samples. Black squares indicate the presence of genetic alterations. Genes are grouped by pathway involvement, denoted by the color-coded bar on the left of the heat map: transcriptional regulation (brown), JAK-STAT (blue), PI3K-AKT (light green), RAS (red), epigenomic (yellow), and Notch signaling (green). The histograms above each column represent the rank-ordered *WNK1* mRNA expression ($\log_2$ [TMP + 1]) per patient sample, with the lowest 10th percentile shown in blue and the top 10th percentile in red. The histogram on the right shows the percentage of each gene mutation across the data set. The bars below the heat map represent clinical characteristics: (a) age group: pediatric (light blue) and adult (brown); and (b) immunophenotype: leukemia subtypes, including early T-cell precursor (ETP), pro-B, pre-B, cortical, and medullary, are color coded. (B) The heat map shows the OncoPrint distribution of genetic alterations across key signaling pathways in 287 pediatric T-ALL samples.¹⁸ Rows represent individual genes, and columns correspond to patient samples. Black squares indicate the presence of genetic alterations. Genes are grouped by pathway involvement, denoted by the color-coded bar on the left of the heat map: transcriptional regulation (brown), JAK-STAT (blue), PI3K-AKT (light green), RAS (red), epigenomic (yellow), and Notch signaling (green). The histograms above each column represent the rank-ordered *WNK1* mRNA expression ($\log_2$ [TMP + 1]) per patient sample, with the lowest 10th percentile shown in blue and the top 10th percentile in red. The histogram on the right shows the percentage of each gene mutation across the data set. The bars below the heat map represent clinical

sets,^{18,21} we demonstrated that *WNK1* is moderately expressed across all major oncogenic subtypes and more highly expressed in the *TAL1/2*-related (TAL-R) subgroups (Figure 2A-D) where *PTEN*/AKT mutations are particularly abundant.²⁷

An outlier analysis of the same set at the 10th or 90th expression percentiles revealed a correlation with *PTEN* mutations (Fisher exact test $P = .027$) and chromosome 6q deletions (del(6q); Fisher exact test $P = .0372$), key aberrancies in poor-prognosis adult T-ALL (Figure 2A-B; supplemental Figure 2B). *PTEN* protein expression is anticorrelated with pWNK-S382, a marker of *WNK1* activation (supplemental Figure 2C). Consequently, loss of *PTEN* causes upregulation of *WNK1* mRNA and protein expression (supplemental Figure 2D), suggesting that *PTEN* may act, in certain T-ALL context, as a negative regulator of the *WNK1* pathway. To support our findings, we also analyzed transcriptional data from a recent study that identified 15 T-ALL subtypes,²³ including *TAL1* $\alpha\beta$ -like and double-positive (DP)-like. The $\alpha\beta$ -like subgroup, enriched for phosphatidylinositol 3-kinase (PI3K) alterations, *PTEN* deletions, and 6q loss, showed higher *WNK1* expression than the DP-like subgroup (Figure 2E; supplemental Figure 2E).

*TAL1*²⁸ binds the promoter region of *WNK1*, which displays enrichment for the histone mark H3K27ac, indicative of active transcription (Figure 2F). In contrast, such enrichment is not observed at the promoters of *WNK2-4* (supplemental Figure 2F-G). Additional coregulatory factors, such as *RUNX1*²⁸ and *TLX3* also bind this region, which may explain why *WNK1* expression is not restricted to the *TAL1* subgroup (Figure 2F; supplemental Figure 2H). Consistently, ectopic modulation of the *TAL1* changed *WNK1* mRNA and protein abundance (supplemental Figure 2I-J) and, as reported previously, chromatin accessibility at the *WNK1* promoter, as shown by assay for transposase-accessible chromatin sequencing (ATAC-seq) (supplemental Figure 2K).²⁹ These findings suggest that the *TAL1* complex actively promotes *WNK1* transcription in T-ALL cells.

Previous studies, in non-T-cell contexts, have shown that AKT-mediated phosphorylation of *WNK1* at threonine 60 (T60)³⁰ enhances its kinase activity.³¹ As such, we profiled 17 AKT-regulated phosphoproteins in 15 T-ALL cell lines and healthy donor T cells using a Luminex-based bead-based assay (Figure 3A; supplemental Table 3.1). Among phosphoproteins, *WNK1* and the proto-oncogene cAMP-response element binding protein (CREB) showed the highest mean fluorescence intensity (Δ MFI) between T-ALL cell lines and normal T cells (Figure 3B). Although normal T cells lacked detectable *WNK1*

phosphorylation, T-ALL samples showed heterogeneous activation (Figure 3C; supplemental Figure 2A). Although T-ALL cell lines reflect genetic heterogeneity, their phosphorylation status may not fully capture the BM microenvironment. To address this, we profiled 16 primary T-ALL samples and found *WNK1* among the most phosphorylated proteins ($n = 17$), suggesting a cell-intrinsic activation mechanism (Figure 3D; supplemental Table 3.2).

Collectively transcriptomic and proteomic data suggest that *WNK1* is not only highly expressed and active but may be particularly relevant in specific subtypes, such as *TAL1/2*-driven T-ALL, where AKT inhibition is relieved due to *PTEN* loss.

Because *WNK1* is highly expressed in thymocytes and is required for progression through differentiation and proliferation in DP thymocytes,³² we anticipated its expression in the human thymus and confirmed this expectation by analyzing single-cell RNA data from extensive thymus atlas data sets (supplemental Figure 3A-B).^{33,34} We found that phospho-*WNK1* were more robustly expressed in clinical T-ALL BM biopsies compared with normal thymus and lymph nodes (Figure 3E; supplemental Figure 3C-D). In line with this observation, we demonstrated that *WNK1* transcripts were significantly higher in *TAL1* primary T-ALL samples than in thymocytes, suggesting *TAL1* oncogenic signals derail physiological *WNK1* control (supplemental Figure 3E).

These results are consistent with the involvement of *WNK1* in T-ALL, particularly in the high-risk *TAL1/2* subgroups.

WNK1 inhibition alters T-ALL proliferation in vitro and preclinical T-ALL models

Given their role in ion homeostasis and blood pressure regulation,^{10,35} WNK kinases have been the focus of ATP-competitive³⁶ and allosteric inhibitor development.^{37,38} WNK463 is an orally potent pan-WNK inhibitor with low nanomolar affinity (K_d: 1nM-9nM) and high kinase selectivity. Due to limited data on the isoform selectivity of this probe, we repeated a *WNK1-4* kinome profiling using 1 μ M and 10 μ M concentrations. Our results indicate that *WNK1* is inhibited under these experimental conditions, further supporting the investigation of WNK463 in our models (supplemental Figure 4A).

We then characterized WNK perturbation in a panel of 14 T-ALL cell lines upon WNK463 treatment. TAL-R cell lines, such as HSB-2, MOLT-16, RPMI-8402, and Jurkat, were more drug-sensitive to WNK463 treatment compared with non-TAL-R cells. 50% inhibitory concentrations for sensitive lines in the low micromoles range: HSB-2: 0.09 $\pm$ 0.2 (from 0.0297 μ M to

Figure 2 (continued) characteristics: (a) del(6q) status: present (green) and absent (white); and (b) transcriptional subtypes, including LMO/LYL1, HOXA, TLX3, TLX1, NKX2-1, LMO1/2, TAL1, TAL2, and unknown are color-coded. (C) Box-and-whisker plot displaying *WNK1* mRNA expression in the TAL-R subset compared with the other T-ALL transcriptional and phenotypic subtypes, including TLX1, TLX3, MLL-T, immature, and unknown. Primary sample data were retrieved from the GSE26713 data set.²² Statistical significance among groups (**** $P \leq .0001$) was determined by 1-way ANOVA using Bonferroni's correction for multiple comparison testing. (D) Box plot shows higher *WNK1* expression in the *TAL1* subset compared with the other T-ALL transcriptional subtypes, including LMO/LYL1, HOXA, TLX3, TLX1, NKX2-1, LMO1/2, TAL2, and unknown. Primary sample data were obtained from T-ALL project data set phs000218.¹⁸ Statistical significance among groups (* $P \leq .05$; ** $P \leq .01$; and ns = not significant) was determined by 1-way ANOVA using Bonferroni's correction for multiple comparison testing. (E) Uniform manifold approximation and projection scatterplot shows the expression of *WNK1* in 1309 primary T-ALL samples.²³ Each subtype is indicated and colored according to the *WNK1* expression represented as z score log₂ FPKM. (F) Chromatin immunoprecipitation sequencing tracks showing *TAL1*, *TLX3*, and *RUNX1* binding and H3K27ac enrichment across the *WNK1* locus in MOLT-3 (GSE59657),²⁴ Jurkat (GSE688976),²⁵ CCRF-CEM (GSE76783),²⁶ MOLT-4 (GSE230594), and DND-41 (GSE199693, GSE230594) T-ALL cell lines. The bottom bar represents the genes (hg19), and yellow highlighting indicates the *WNK1* promoter region.

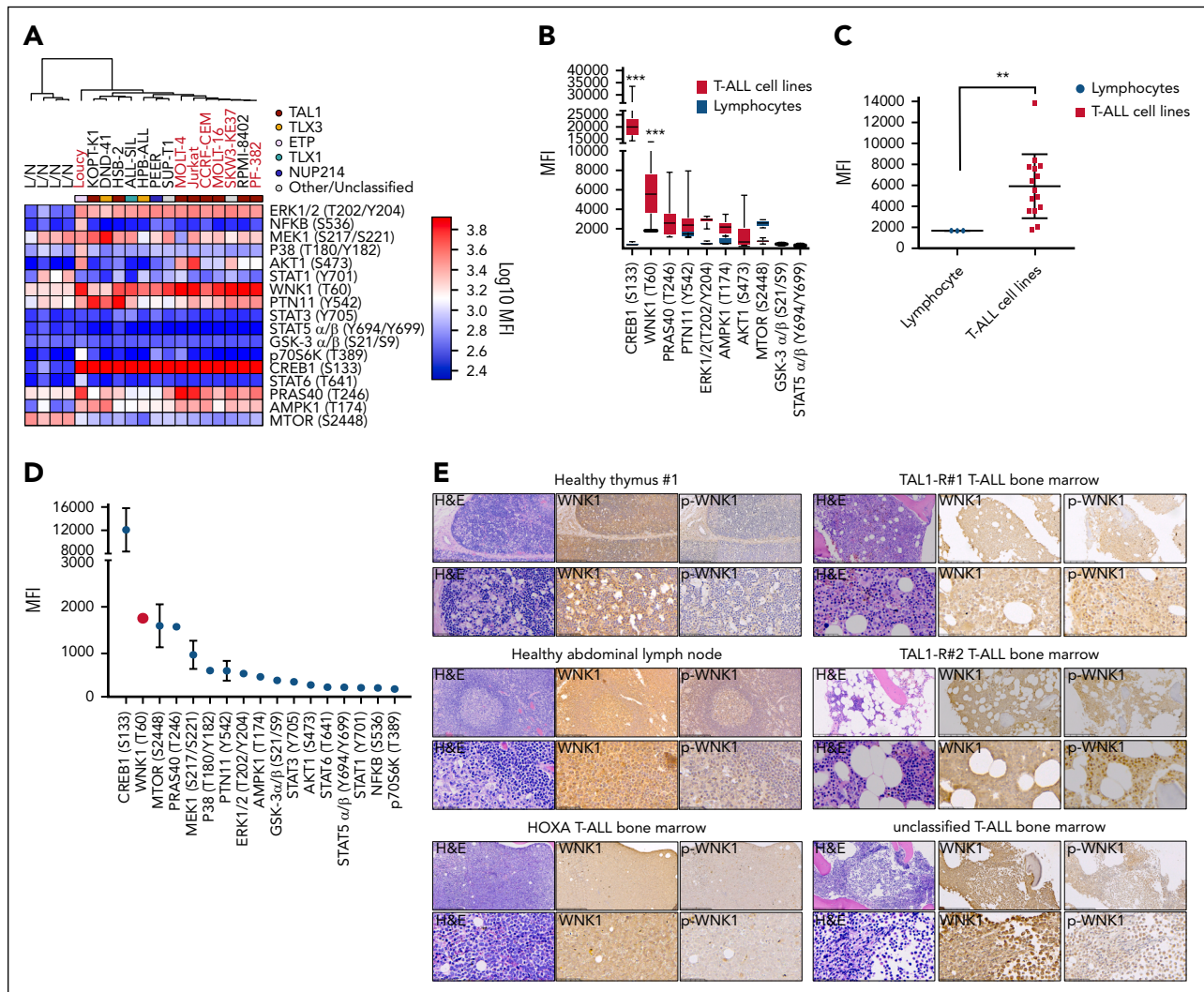


Figure 3. Aberrant phosphorylation and activation of WNK1 in T-ALL. (A) Heat map displaying the $\log_{10}$ -transformed absolute median fluorescent intensity (MFI) values for the indicated phosphorylated kinases. Each column represents the mean of 3 replicates for T-ALL cell lines ($n = 15$) and lymphocytes ($n = 4$), whereas rows represent individual phosphorylated kinases. The color scale indicates the relative expression level of each kinase. Unsupervised clustering was performed using Pearson correlation coefficients. The T-ALL cell lines highlighted in red are *PTEN* deleted. (B) Box-and-whisker plot comparison of phosphorylated protein expression in T-ALL cell lines ($n = 15$) compared with lymphocytes ($n = 4$). The y-axis shows a MFI in T-ALL (red) and lymphocytes (black). Error bars indicate the mean $\pm$ standard deviation (SD) of 15 cell lines and 4 lymphocytes measured in technical replicates. Statistical significance among groups ($***P \leq .001$) was assessed using the Mann-Whitney U test. (C) Scatterplot comparison of the MFI of activated p-WNK1-T60 in a panel of T-ALL cell lines ($n = 15$) and lymphocytes ($n = 3$). Statistical significance among groups ($**P \leq .01$) was determined by Mann-Whitney U test. (D) Scatter dot plot shows MFI of phosphorylated activated kinases in T-ALL primary samples ($n = 16$). A red dot represents p-WNK1-T60. Each dot represents the mean $\pm$ standard error of the mean (SEM) of 3 replicates per T-ALL primary sample. (E) WNK1 and phospho-WNK1-T60 expression (brownish) in human thymus, lymph nodes, and BM leukemia cells. Formalin-fixed, paraffin-embedded (FFPE) tissue sections were stained with anti-WNK1 and anti-p-WNK1-T60 antibodies, revealed by immunoperoxidase. Scale bars represent 250 μm and 50 μm in top and bottom images, respectively, for each staining.

0.253 μM); Jurkat: 1.19 $\pm$ 0.02 (from 1.101 μM to 1.288 μM); RPMI-8402: 0.61 $\pm$ 0.02 (from 0.554 μM to 0.668 μM); MOLT-16: 0.36 $\pm$ 0.02 (from 0.3124 μM to 0.4188 μM ; Figure 4A). No significant correlation was observed between antibody-dependent P-WNK1 levels, as assessed in Figure 3A, and sensitivity to WNK463 (supplemental Figure 4B). Multiple measurements of cell death and apoptosis confirmed a greater WNK463 effect in the TAL-R T-ALL cell lines subtype (Figure 4B-D; supplemental Figure 4C-G). Treatment with WNK-IN-11 treatment, an allosteric non-ATP-competitive WNK inhibitor,³⁷ was associated with a similar impact on T-ALL proliferation, induction of cell death, and cleaved poly(ADP-ribose) polymerase (PARP) (supplemental Figure 4H-K). WNK463 combined with standard-of-care agents showed

synergistic or additive effects in T-ALL cells (supplemental Figure 5A-B).

To ascertain whether the effects seen on cell viability upon WNK463 suppression were, at least in part, mediated by on-target repression of WNK1, we overexpressed in 293T cells a synthetic plasmid expressing Myc-tagged wild-type short form WNK1-(1-491) or the double mutant S378A/S382A (SSAA) in the WNK1 catalytic domain.⁸ This mutation is predicted to block WNK463 binding to WNK1 and prevent the phosphorylation of its target SPAK and OSR1.⁹ As shown in supplemental Figure 4L-M, overexpression of the WNK1 mutant isoform rescues 293T cells from WNK463-induced mediated suppression, as indicated by the lack of a decrease

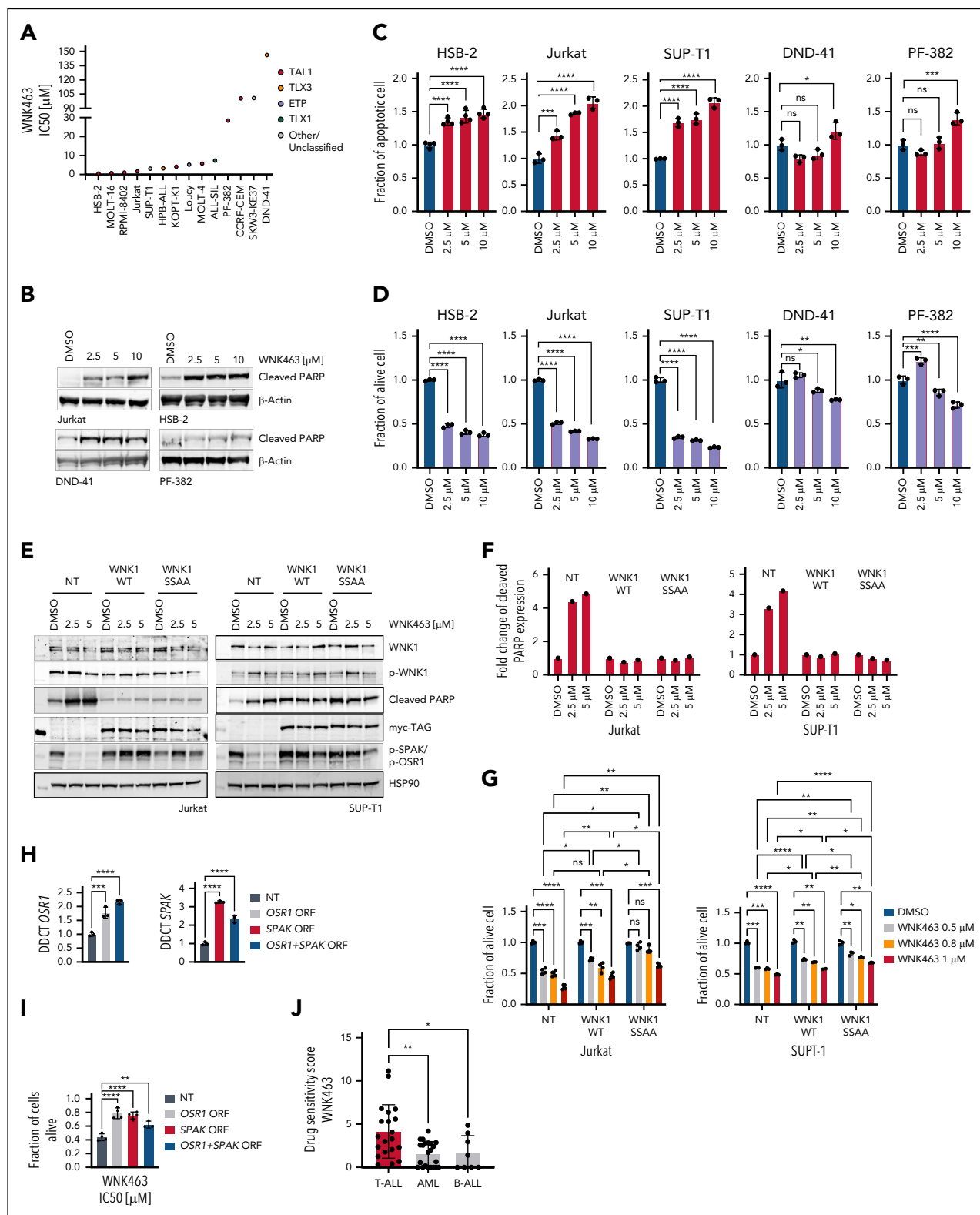


Figure 4. WNK463 inhibits T-ALL cell growth in vitro. (A) The dot plot shows the effect of WNK463 on cell viability after 72 hours of treatment in T-ALL cell lines (n = 14). 50% inhibitory concentration values were determined after 72 hours of treatment with WNK463 at 10 increasing concentrations (0.5 μM–20 μM) using an ATP-based luminescence viability assay. (B) Western blot showing cleaved-PARP protein expression in T-ALL cell lines after 24 hours of treatment with WNK463 or dimethyl sulfoxide (DMSO). β-Actin was used as a loading control. (C) Effects of WNK463 treatment on cell apoptotic fraction in HSB-2, Jurkat, SUP-T1, DND-41, and PF-382 after 48 hours of drug exposure. Statistical significance among groups (*P ≤ .05; ***P ≤ .001; ****P ≤ .0001; ns = not significant) was determined by 1-way ANOVA using Dunnett's correction for multiple comparison testing. Data are presented as mean ± SD of 3 replicates. (D) Effects of WNK463 treatment on cell viability in HSB-2, Jurkat, SUP-T1, DND-41, and PF-382 after 72 hours of drug exposure. Statistical significance among groups (*P ≤ .05; **P ≤ .01; ***P ≤ .001; ****P ≤ .0001; ns = not significant) was determined by 1-way ANOVA using Dunnett's correction for multiple comparison testing. Data are presented as mean ± SD of 3 replicates. (E) Western blot showing WNK1, p-WNK1-T60,

in the WNK1 targets, p-SPAK/p-OSR1. T-ALL cell lines overexpressing S378A/S382A (SSAA) mutations (Figure 4E) exhibit reduced sensitivity to WNK463 treatment as do cells overexpressing wild-type WNK1 compared with T-ALL transduced with an empty vector (NT; Figure 4F-G), suggesting that the impact of the molecule on T-ALL viability is mediated by WNK1 pathway inhibition. Consistently, overexpression of the WNK1 downstream kinases *SPAK* and *OSR1* attenuated the effect of WNK463 in T-ALL cells (Figure 4H-I). We next compared the response to WNK1 inhibitors across different types of sample from patients with acute leukemia. By a differential drug-sensitivity score, a robust quantitative measurement that summarizes cellular response to a drug across a range of concentrations, WNK463 displayed a varying degree of responses as seen in cell lines (Figure 4A), although with a more evident effect in T-ALL compared with its impact on other acute malignancies, such as AML or B-cell acute lymphoblastic leukemia (Figure 4J).

To evaluate WNK463 in a clinically relevant model, we established a PDLX TAL-R model by serially transplanting *STIL::TAL1*⁺ leukemia cells into NSG mice for in vivo and ex vivo treatments. We first confirmed that lymphoblasts from the TAL-R model exhibited a dose-dependent increase in apoptotic death upon ex vivo WNK463 treatment (supplemental Figure 6A), which correlated with a reduction in P-WNK1 and its downstream targets, p-SPAK/p-OSR1 (supplemental Figure 6B). Ten mice per group were treated in vivo with WNK463 at 1.5 mg/kg per day for 26 days (Figure 5A). By day 10, we observed a significant reduction in human CD45⁺ cells in the PB of treated mice (Figure 5B). At the end point for the control group, we sacrificed 4 mice from each group for efficacy studies, revealing decreased leukemic cell infiltration in the BM and spleen ($P = .0286$; Figure 5C). This effect was marked by reduced spleen size (Figure 5D-E) and pharmacodynamically with a decrease in WNK1 and SPAK phosphorylation (Figure 5F-G; supplemental Figure 6C). Importantly, we failed to recapitulate these data in an unclassified non-TAL1 T-ALL PDLX model (Figure 5B-E), although comparison between WNK1 expression levels was not performed before treatment. The remaining *STIL::TAL1* mice were monitored for survival analysis, where WNK463-treated mice demonstrated a significant survival advantage compared with the vehicle control ($P = .0009$, log-rank Mantel-Cox test; Figure 5H). WNK463 treatment exhibited excellent tolerability, with no significant drug-related toxicity, including no weight loss (supplemental Figure 6D-F), secondary cytopenias (supplemental Figure 6E-G), or organ defects (Figure 5I; supplemental Figure 6H-I).

These data position WNK1 as a candidate therapeutic target in T-ALL and demonstrate that WNK463 reduces T-ALL growth in xenograft TAL-R models, supporting further preclinical optimization of WNK1 chemical probes.

WNK1 accelerates leukemia establishment in TAL-R PDLX models

To provide further evidence that repression of WNK1 provides therapeutic benefit in T-ALL, we turned to genetic approaches with RNA interference. We transduced Jurkat and HSB-2 cells with shRNAs targeting WNK1 (Figure 6A) and assessed the impact of WNK1 loss on cell proliferation and on the downstream kinase substrate WNK1 targets, SPAK and OSR1³⁹ (Figure 6A). WNK1 loss preferentially impaired the viability of TAL1 T-ALL cells in vitro (Figure 6B-C; supplemental Figure 7A-B), confirming WNK1 dependency in this subgroup. Furthermore, custom-designed sgRNAs Cas9-mediated targeting the WNK1 kinase pocket reduced WNK1 expression (Figure 6D) and decreased cell viability (Figure 6E-F) compared with non-targeting sgRNA controls (NT), suggesting that TAL-R T-ALL relies on WNK1 for survival.

To investigate the role of WNK1 in leukemia establishment, we developed a PDLX model in which *STIL::TAL1*⁺ primary leukemia cells were transduced with a tetracycline-inducible shRNA targeting WNK1. Tetracycline treatment at 2 mg/mL efficiently suppressed WNK1 and its downstream targets SPAK and OSR1 (supplemental Figure 7C). Immunodeficient NSG mice then received cells through transplant, with leukemia establishment monitored via mCherry expression by flow cytometry (Figure 6G). We tested whether WNK1 silencing in vivo delays leukemia cell engraftment. As shown in Figure 6H, WNK1 loss significantly reduced leukemia progression in PB and the number of *STIL::TAL1*⁺ cells in the BM (Figure 6I; supplemental Figure 7D). A more significant effect was seen in leukemic cells infiltrating secondary lymphoid organs such as the spleen (Figure 6J-L; supplemental Figure 7E-H), in line with the idea that WNK1 may orchestrate T-cell homing to lymphoid organs.⁴⁰ A reduction in leukemia-cycling BM cells was observed only in doxycycline-treated mice, as assessed by Ki-67 quantification (Figure 6M-N; supplemental Figure 7I). This was linked to WNK1 inactivation, evidenced by the absence of WNK1 expression in BM and spleen leukemia cells from sacrificed mice (supplemental Figure 7J-K). Finally, *STIL::TAL1* mice lacking WNK1 expression showed a reduction in the WNK1 pathway activation, as shown by organ P-SPAK and P-WNK1 expression (Figure 6O-Q), and therefore a prolonged survival

Figure 4 (continued) p-SPAK/p-OSR1, myc-TAG, and cleaved-PARP protein expression in myc-tagged WNK1-(1-491) wild-type (WT) or the double mutant S378A/S382A (SSAA) in Jurkat and SUP-T1 cell lines after 24 hours of treatment with WNK463 or DMSO. Lysates were stained with the anti-WNK1 total, anti-phospho-WNK1-T60, anti-phospho-SPAK (Ser373)/phospho-OSR1 (Ser325), anti-myc-TAG, and anti-cleaved PARP total antibodies. HSP90 was used as a loading control. NT = nontargeting, WT = WNK1-(1-491) WT and SSAA = double mutant S378A/S382A. (F) Histograms showing densitometric quantification of cleaved-PARP protein after 24 hours of WNK463 treatment in Jurkat and SUP-T1 cell lines overexpressing myc-tagged WNK1-(1-491) WT or the double mutant S378A/S382A (SSAA). The relative intensity of cleaved-PARP was normalized for the levels of HSP90 and expressed as fold change relative to DMSO. NT = nontargeting, WT = WNK1-(1-491) wild type and SSAA = double mutant S378A/S382A. (G) Effects of WNK463 treatment on cell viability in SUP-T1 and Jurkat overexpressing WNK1 WT, WNK1 SSAA, or NT after 72 hours of drug exposure. Statistical significance among groups (* $P \leq .05$; ** $P \leq .01$; *** $P \leq .001$; **** $P \leq .0001$) was determined by 2-way ANOVA using Tukey's multiple comparison testing. Data are presented as mean $\pm$ SD of 3 replicates. NT = nontargeting, WT = WNK1-(1-491) wild type and SSAA = double mutant S378A/S382A. (H) Expression of *OSR1* and *SPAK* mRNA relative to the control gene *ACTB* ($\Delta\Delta CT$) in Jurkat cells 6 days after ORF transduction. Data are presented as mean $\pm$ SD ($n = 3$). Statistical significance among groups (*** $P \leq .001$; **** $P \leq .0001$) was determined by a 1-way ANOVA using Dunnett's correction for multiple comparison testing. NT = nontargeting. (I) Effect of SPAK and OSR1 overexpression in Jurkat cell lines at 6 days ORF transduction on cell viability. Error bars denote the mean $\pm$ SD of a minimum of 3 biological replicates. Statistical significance among groups (** $P \leq .01$; *** $P \leq .001$; **** $P \leq .0001$) was determined by 1-way ANOVA using Dunnett's correction for multiple comparison testing. NT = nontargeting. (J) Effects of WNK463 in T-ALL ($n = 20$), AML ($n = 22$), and B-ALL ($n = 8$) primary samples. The log-transformed dose-response data's drug-sensitivity score model is depicted. Statistical significance among groups (* $P \leq .05$; ** $P \leq .01$) was determined by a 1-way ANOVA using Dunnett's correction for multiple comparison testing.

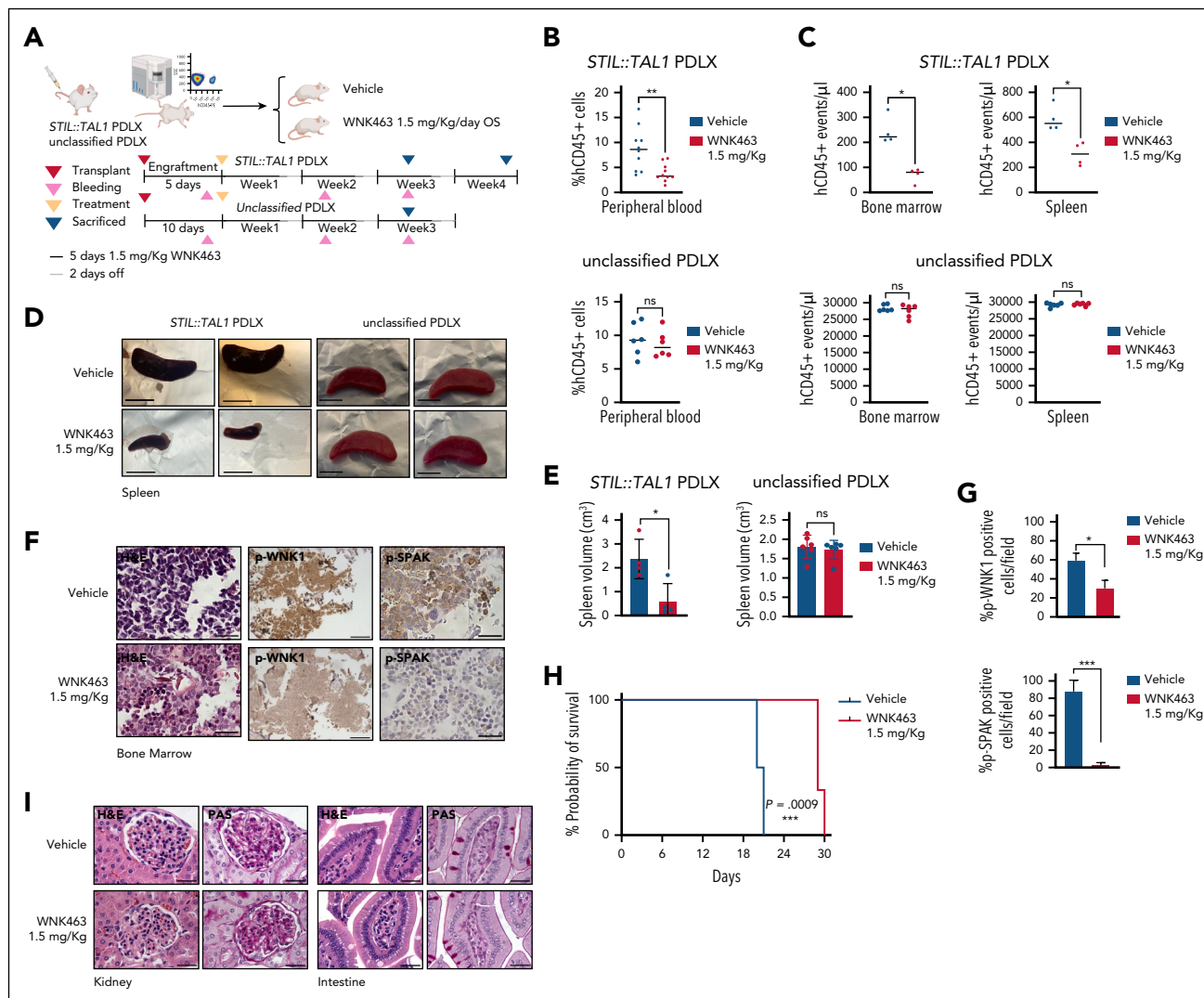


Figure 5. WNK463 inhibits T-ALL cell growth in vivo preclinical models. (A) Schematic representation of WNK463 in vivo efficacy. Mice (STIL::TAL1 PDLX and unclassified) were treated with vehicle, or 1.5 mg/kg WNK463 (5 d/wk) for 4 or 3 weeks, respectively. Figure partially created with [biorender.com](#). (B) Effect of WNK463 treatment on hCD45⁺ cells in PB. Data are presented as a percentage relative to hCD45⁺ cells. Mice were treated with vehicle (STIL::TAL1 PDLX n = 10 and unclassified n = 6, blue) or 1.5 mg/kg WNK463 (STIL::TAL1 PDLX n = 10 and unclassified n = 6, red) for 10 days. Statistical significance among groups (**P ≤ .01) was determined by Mann-Whitney U test. (C) Impact of WNK463 on hCD45⁺ cell counts in BM and spleen. Mice were treated with vehicle (STIL::TAL1 PDLX n = 4 and unclassified n = 6, blue) or 1.5 mg/kg WNK463 (STIL::TAL1 PDLX n = 4 and unclassified n = 6, red) for 17 days. The number of hCD45⁺ events per microliter is shown. Statistical significance among groups (*P ≤ .05) was determined by Mann-Whitney U test. (D) Representative spleen images of mice treated with 1.5 mg/kg WNK463 (n = 2) or vehicle (n = 2) after 17 days. Scale bar, 1 cm. (E) Impact of WNK463 on spleen size. Mice were treated with vehicle (STIL::TAL1 PDLX n = 4 and unclassified n = 6, blue) or 1.5 mg/kg WNK463 (STIL::TAL1 PDLX n = 4 and unclassified n = 6, red) for 17 days. The spleen volume (cubic centimeter) is shown. Statistical significance among groups (*P ≤ .05) was determined by Mann-Whitney U test. ns = not significant. (F) Phospho-WNK1-T60 and phospho-SPAK-S373 expression (brownish) in BM fragments from PDLX mice after 17 days of 1.5 mg/kg WNK463 (5 days a week) or vehicle. FFPE sections were stained with anti-p-WNK1-T60 and anti-phospho-SPAK-S373 antibodies revealed by immunoperoxidase. Scale bar, 30 μm. (G) Histograms indicate the mean ± SD of the percentage of the p-WNK1-T60 and SPAK-S373⁺ cells in FFPE tissue sections after 17 days of 1.5 mg/kg WNK463 (5 days a week) or vehicle. Statistical significance (*P ≤ .05, ***P ≤ .001) was determined by Mann-Whitney U test. (H) Kaplan-Meier survival plot showing the overall survival (OS) of STIL::TAL1 PDLX T-ALL mice treated with vehicle (10% DMSO + 90% corn oil) or WNK463 (1.5 mg/kg for 26 days; ***P ≤ .001). (I) Representative hematoxylin and eosin (H&E) and Periodic Acid-Schiff (PAS) staining of kidney and intestine section from STIL::TAL1 PDLX mice after 17 days of 1.5 mg/kg WNK463 (5 days a week) or vehicle. No significant morphological alterations, structural damage, or organ toxicity were observed in WNK463-treated mice compared with vehicle controls. Scale bar, 30 μm.

(P < .0001, log-rank Mantel-Cox test) compared with NT control (Figure 6R).

Collectively, these data suggest that WNK1 plays a role in the establishment and progression of T-ALL, including TAL1/2-associated T-ALL, and that its downregulation may reduce the risk of secondary organs involvement, a common finding at clinical relapse.

WNK1 inhibition disrupts phases of the cell cycle and mitosis

To investigate the pathways affected by WNK1 suppression, we analyzed the transcriptional consequences of WNK1 loss in RNA isolated from Jurkat cells expressing either nontargeting control shRNA (NT) or 2 different WNK1-targeting shRNAs #919 and #920. We detected changes in RNA quantity with >500 transcripts changing from log₂-fold ≥0.5 and P ≤ .01

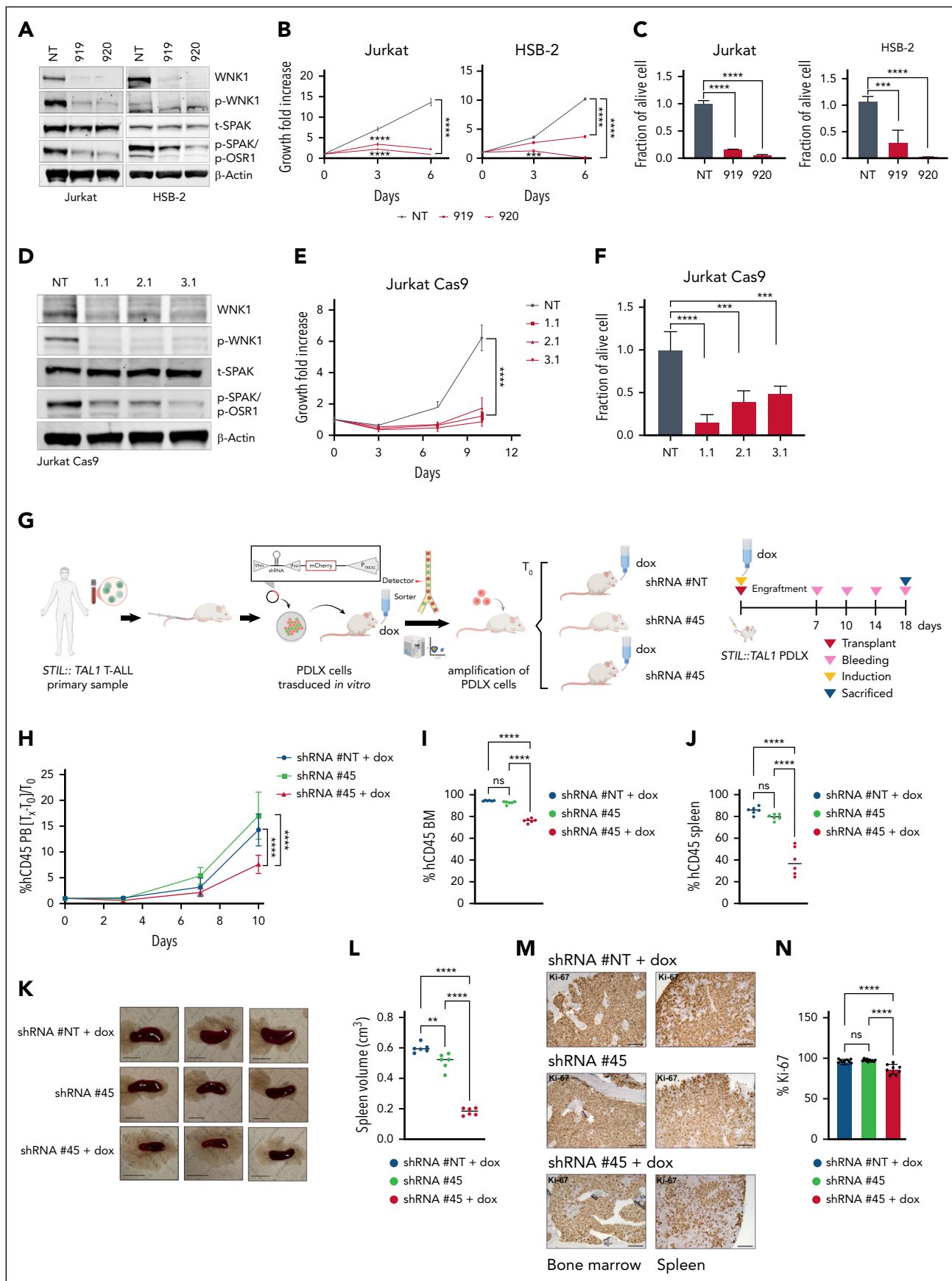


Figure 6. Genetic loss of *WNK1* impairs T-ALL proliferation in vitro and in vivo. (A) Western blot showing WNK1, p-WNK1-T60, SPAK, and p-SPAK/p-OSR1 protein expression in Jurkat (left) and HSB-2 (right) cell lines, 3 days after selection after shRNA transduction. Lysates were stained with the anti-WNK1 total, anti-phospho-WNK1-T60, anti-phospho-SPAK (Ser373)/phospho-OSR1 (Ser325), and anti-SPAK total antibodies. β -Actin was used as a loading control. NT = nontargeting, #919 and

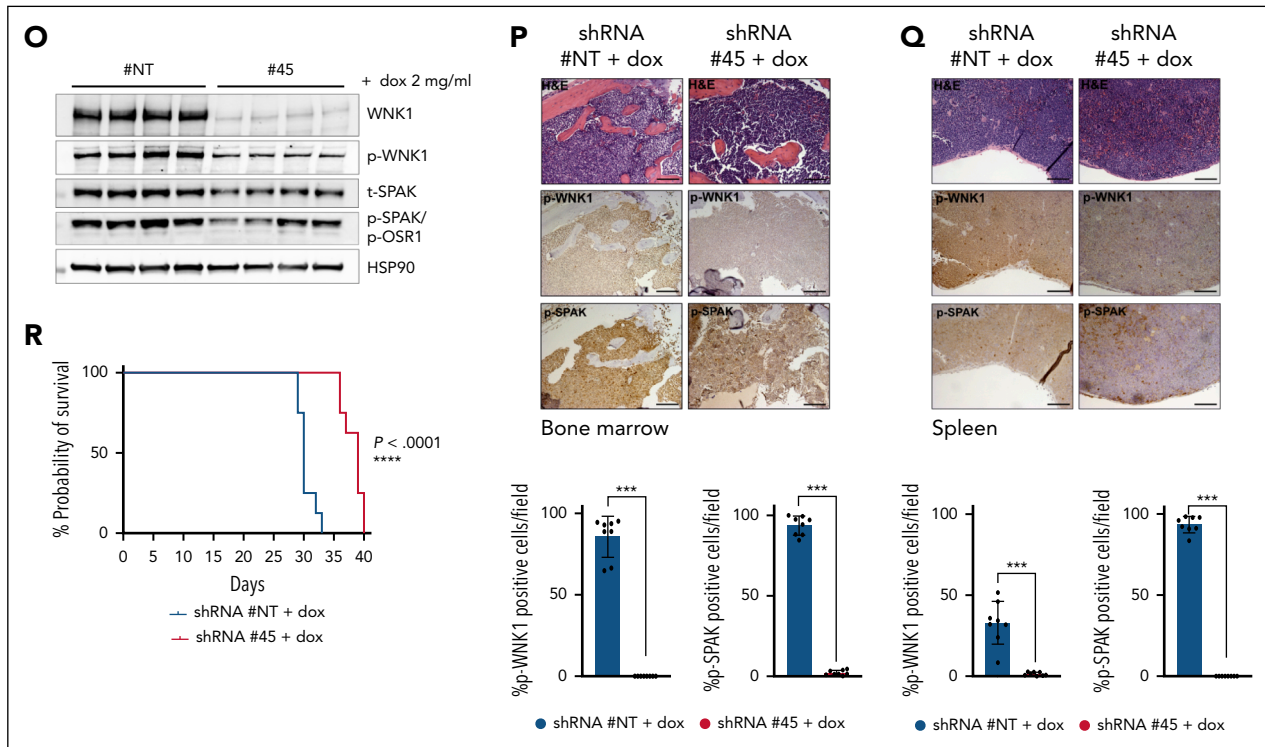


Figure 6 (continued) #920 = shRNA directed against *WNK1*. (B) Effect of *WNK1* loss in Jurkat (left) and HSB-2 (right) cell lines at 3 or 6 days after shRNA selection on cell proliferation. Error bars denote the mean $\pm$ SD of a minimum of 3 biological replicates. Statistical significance among groups ($***P \leq .001$; $****P \leq .0001$) was determined by 2-way ANOVA using Bonferroni's correction for multiple comparison testing. NT = nontargeting, #919 and #920 = shRNA directed against *WNK1*. (C) Effect of *WNK1* loss in Jurkat (left) and HSB-2 (right) cell lines at 6 days after shRNA selection on cell viability. Error bars denote the mean $\pm$ SD of a minimum of 3 biological replicates. Statistical significance among groups ($***P \leq .001$; $****P \leq .0001$) was determined by 1-way ANOVA using Dunnett's correction for multiple comparison testing. NT = nontargeting, #919 and #920 = shRNA directed against *WNK1*. (D) Western blot showing *WNK1*, p-*WNK1*-T60, *SPAK*, and p-*SPAK*/p-*OSR1* protein expression in Jurkat-Cas9 cells, 7 days after selection after sgRNA transduction. Lysates were stained with the anti-*WNK1* total, anti-phospho *WNK1*-T60, anti-phospho-*SPAK* (Ser373)/phospho-*OSR1* (Ser325), and anti-*SPAK* total antibodies. β -Actin was used as a loading control. NT = nontargeting, #1.1, #2.1, and #3.1 = sgRNA directed against *WNK1*. (E) Effect of *WNK1* loss on proliferation of Jurkat-Cas9 cells at 3, 7, or 10 days after sgRNA selection. Error bars denote the mean $\pm$ SD of a minimum of 3 biological replicates. Statistical significance among groups ($***P \leq .001$; $****P \leq .0001$) was determined by 2-way ANOVA using Bonferroni's correction for multiple comparison testing. NT = nontargeting, #1.1, #2.1, and #3.1 = sgRNA directed against *WNK1*. (F) Effect of *WNK1* loss on viability of Jurkat-Cas9 cells 7 days after sgRNA selection. Error bars denote the mean $\pm$ SD of a minimum of 3 biological replicates. Statistical significance among groups ($***P \leq .001$; $****P \leq .0001$) was determined by 1-way ANOVA using Dunnett's correction for multiple comparison testing. NT = nontargeting, #1.1, #2.1, and #3.1 = sgRNA directed against *WNK1*. (G) Schematic representation of the leukemia establishment in vivo study to establish leukemia. *STIL::TAL1* T-ALL primary samples were serially transplanted to generate PDLX models. PDLX cells were lentivirally transduced with a tetracycline-inducible construct targeting *WNK1* in vitro and subsequently transplanted in mice. Mice were exposed to doxycycline hyclate treatment (2 mg/mL in water) 10 days before end point to induce the mCherry-expressing plasmids in the cells carrying shRNA #45 or shRNA #NT, enriched using flow cytometry by gating on the expressed fluorochrome and sorted by fluorescence-activated cell sorting, to be subsequently retransplanted in mice. Mice (*STIL::TAL1* PDLX), carrying shRNA #45 ($n = 6$) and shRNA #NT ($n = 6$), were treated with 2 mg/mL doxycycline for 18 days. A control group (shRNA #45, $n = 6$) was not exposed to doxycycline treatment to prevent the induction of *WNK1* knockdown. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. Figure partially created with [biorender.com](https://www.biorender.com). (H) Percentage of hCD45⁺ cells expressed as the difference between PB leukemic cells at days 3, 7, and 10, and T0 (1 week after injection), normalized for T0. Statistical significance ($****P \leq .0001$) was determined by 2-way ANOVA using Tukey's multiple comparison testing. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (I) Effect of *WNK1* loss on hCD45⁺ cells in BM. Data are presented as a percentage relative to hCD45⁺ cells. Statistical significance ($****P \leq .0001$; ns = not significant) was determined by 1-way ANOVA using Tukey's multiple comparison testing. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (J) Effect of *WNK1* loss on the percentage of hCD45⁺ cells in the spleen. Statistical significance ($****P \leq .0001$) was determined by 1-way ANOVA using Tukey's multiple comparison testing. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (K) Representative spleen images of mice after 18 days of *WNK1* knockdown shRNA #45 induced by doxycycline 2 mg/mL or control group (shRNA #NT and shRNA #45 without doxycycline). Scale bar, 1 cm. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (L) Effect of *WNK1* loss on spleen volume (cubic centimeter) of *STIL::TAL1* PDLX mice after 18 days of *WNK1* knockdown shRNA #45 induced by doxycycline 2 mg/mL or control group (shRNA #NT and shRNA #45 without doxycycline). Statistical significance ($**P \leq .01$; $****P \leq .0001$) was determined by 1-way ANOVA using Tukey's multiple comparison testing. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (M) Ki-67 expression (brownish) in BM and spleen of mice (*STIL::TAL1* PDLX) after 18 days of *WNK1* knockdown shRNA #45 induced by doxycycline 2 mg/mL ($n = 6$) or control group (shRNA #NT, $n = 6$ and shRNA #45 without doxycycline, $n = 6$). FFPE tissue sections were stained with anti-Ki-67 antibody, revealed by immunoperoxidase. Scale bar, 100 μ m. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (N) Histograms indicate the mean $\pm$ SD of the percentage of the Ki-67⁺ cells in FFPE BM tissue after 18 days of *WNK1* knockdown shRNA #45 induced by doxycycline 2 mg/mL ($n = 6$) or control group (shRNA #NT, $n = 6$ and shRNA #45 without doxycycline, $n = 6$). Statistical significance ($****P \leq .0001$) was determined by 1-way ANOVA using Tukey's multiple comparison testing. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (O) Western blot showing *WNK1*, p-*WNK1*-T60, *SPAK*, and p-*SPAK*/p-*OSR1* protein expression in mice spleen (*STIL::TAL1* PDLX) after 40 days of *WNK1* knockdown shRNA #45 induced by doxycycline 2 mg/mL or control group (shRNA #NT). Lysates were stained with the anti-*WNK1* total, anti-phospho *WNK1*-T60, anti-phospho-*SPAK* (Ser373)/phospho-*OSR1* (Ser325), and anti-*SPAK* total antibodies. HSP90 was used as a loading control. #NT = nontargeting, #45 = shRNA directed against *WNK1*, dox = doxycycline. (P-Q) Immunohistochemical staining for phospho-*WNK1* (T60) and phospho-*SPAK* (S373) (brown) in BM (P) and spleen (Q) of *STIL::TAL1* PDLX mice after 40 days of doxycycline-induced *WNK1* knockdown (shRNA #45, 2 mg/mL dox; $n = 8$) or control mice (shRNA #NT, $n = 8$). FFPE tissue sections were stained with anti-phospho-*WNK1* (T60) and anti-phospho-*SPAK* (S373) antibodies and revealed by immunoperoxidase. Scale bar, 100 μ m. Bottom: histograms showing the mean $\pm$ SD percentage of p-*WNK1* (T60)- and p-*SPAK* (S373)-positive cells. Statistical significance ($***P \leq .001$) was determined using the Mann-Whitney *U* test. #NT = nontargeting control; #45 = shRNA targeting *WNK1*; dox = doxycycline. (R) Kaplan-Meier survival plot showing the OS of *STIL::TAL1* PDLX T-ALL mice after 40 days of doxycycline-induced *WNK1* knockdown (shRNA #45, 2 mg/mL dox; $n = 8$) or control mice (shRNA #NT, $n = 8$). $****P \leq .0001$. #NT = nontargeting control; #45 = shRNA targeting *WNK1*; dox = doxycycline.

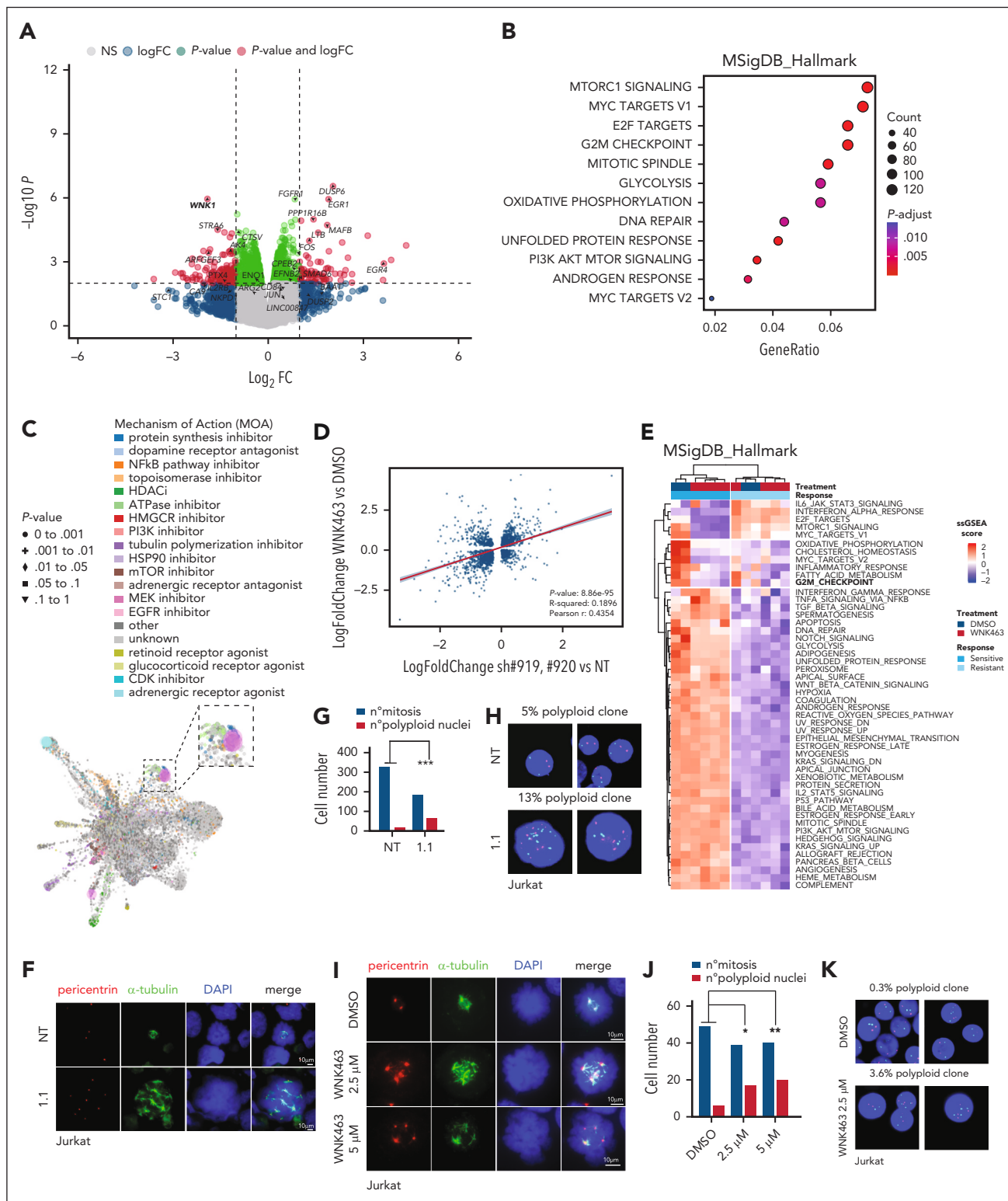


Figure 7. Loss of WNK1 modulates mitosis and cell cycle dynamics. (A) Volcano plot of RNA sequencing expression plot following *WNK1* suppression by knockdown with a *WNK1*-directed shRNA (#919 and #920) or nontargeting control in the Jurkat cell line. Differentially expressed genes with $P < .01$ and $\text{LogFC} \pm 1$ are depicted in magenta. (B) Dot plot of gene set enrichment analysis results illustrating Molecular Signature Database (MSigDB) biological processes associated with *WNK1* loss after shRNA transduction (#919 and #920) in Jurkat cell lines. Gene count refers to the number of genes associated with each (MSigDB) biological process. Gene ratio (x-axis) is the percentage of genes significantly correlated with *WNK1* loss from the total number of genes associated with that process. Dot size is proportional to the number of matched genes within each pathway, whereas the color indicates the adjusted P value for each pathway. (C) L1000FWD fireworks visualization of drug-induced signatures mimicking and reversing the differential gene expression signature generated from *WNK1* shRNA #919 and #920 or NT in Jurkat cell lines. Aquamarine circles indicate small molecules causing a reverse signature *WNK1* "off." Light violet circles indicate small molecules mimicking a signature *WNK1* "off." Protein synthesis and topoisomerase inhibitors are indicated in blue and light orange overlapping with light violet circles. (D) Scatterplot showing significant gene expression changes in sh#919 and sh#920 vs NT or treated

(Figure 7A). *WNK1* scored among the most differentially downregulated genes (supplemental Table 4.1). Gene set enrichment analysis⁴¹ of the significantly differentially expressed genes revealed that alterations in gene sets associated with Myc, MTORC1, and E2F targets (Figure 7B; supplemental Table 4.2), pathways that are commonly downregulated when T-ALL lymphoblast proliferation is inhibited. However, in addition, the G2/M checkpoint and mitotic spindle control (Figure 7B) scored strongly with *WNK1* repression. We next projected marker genes of *WNK1* “on” and “off” into the Library of Integrated Network-Based Cellular Signatures⁴² and demonstrated that the *WNK1* knockdown signature mimics the effects of small molecule belonging to different compound classes but with a reported effect on cell cycle inhibition (Figure 7C; supplemental Table 4.3), such as the topoisomerase I inhibitor narciclasine⁴³ and the protein translational inhibitor V4877,⁴⁴ further supporting the mechanistic contribution of *WNK1* to cellular division and proliferation. In parallel, we assessed the consequence of *WNK1* loss in both sensitive and resistant cell lines (supplemental Figure 8A-B).

Next, we analyzed the transcriptional consequences of WNK463 treatment in *WNK1*-sensitive cells and identified 3788 differentially expressed genes ($\log_2$ -fold change ≥ 0.5 , $P \leq .01$; supplemental Figure 8C; supplemental Table 4.4). Although WNK463 functions as a pan-WNK inhibitor, the genetic and small-molecule perturbation signatures significantly overlap (Figure 7D; supplemental Table 4.5).

As we observed upon *WNK1* genetic repression by shRNA (Figure 7B), drug treatment also led to modulation of the G2/M checkpoint signature in *WNK1*-sensitive cells (supplemental Figure 7D; supplemental Table 4.6; Figure 7E), with differences in the transcriptional profiles of sensitive vs resistant cells (supplemental Figure 8E). Single-sample gene set enrichment analysis further confirmed that WNK463 treatment modulates the G2/M checkpoint signature in sensitive cells compared with resistant ones (Figure 7E), suggesting a role for *WNK1* in regulating cell cycle progression and mitotic entry. Consistently, in TAL-R cell lines, *WNK1* loss promoted polyploidy and mitotic spindle defects (Figure 7F-H; supplemental Figure 8F-H). Exposure to WNK463 induced morphometric changes, including incomplete cell division and aberrant chromosome segregation, attributable to defects in the mitotic spindle and abscission (Figure 7I-K), confirming observations previously reported in HeLa cells.⁴⁵ Importantly, overexpression of the S378A/S382A (SSAA) *WNK1* mutant abolishes this response in Jurkat cells (supplemental Figure 8I-J), confirming that it is dependent on *WNK1* modulation.

These findings indicate that *WNK1* is crucial for cell cycle progression in T-ALL, and that its aberrant activation may be exploited as a therapeutic vulnerability.

Discussion

At first glance, T-ALL might appear to be a disease with a unique genetic driver lesion, ideal for 1-fits-all targeted approaches. In fact, most cases involve an oncogenic event that directly or indirectly affects *NOTCH1* signaling, such as through gain-of-function mutations.⁴⁶ However, attempts to target *NOTCH1* for clinical benefit have largely been unsuccessful, and outcomes in the relapsed/refractory setting remain poor.⁴⁷⁻⁴⁹

Among more druggable and broadly expressed targets, only a few kinases, such as Janus kinase (JAK) and lymphocyte-specific protein tyrosine kinase (LCK), have shown therapeutic relevance. JAK2 alterations (eg, single nucleotide variants or *ETV6::JAK2* translocations) are rare in T-ALL (<1%), whereas mutations in *JAK1*, *JAK3*, or the *IL7R-JAK-STAT* pathway occur in >10% of patients, supporting clinical investigation.^{50,51} Ruxolitinib and other JAK inhibitors showed preclinical activity, with early but limited efficacy in trials (ClinicalTrials.gov identifiers: NCT05745714 and NCT03117751), and off-trial use.⁵² LCK targeting, instead, relies on aberrant activation rather than mutation frequency. Dasatinib showed *ex vivo* activity in 30% to 40% of patients with T-ALL,⁵³ later linked to LCK inhibition via *in silico* analysis.⁵⁴ LCK activation defines distinct T-ALL subgroups,⁵⁵ broadening dasatinib's potential use beyond *NUP214::ABL1*⁺ cases.⁵⁶⁻⁵⁸

WNK1 is generally not mutated in T-ALL. We demonstrated that it is highly expressed in T-ALL, especially within the poor-prognosis TAL oncogenic subset, which accounts for >30% of the T-ALL cases, and is characterized by the overexpression of *TAL1/2* genes due to the presence of fusions such as *STIL::TAL1*, *TAL2*, or the generation of a *de novo TAL1* enhancer.²⁴ Within the TAL subset, there is a high frequency of inactivating monoallelic *PTEN* mutations, microdeletions, or 6q deletions, all of which are associated with poor prognosis.^{59,60}

Because *TAL1/2* subgroups correspond to a late cortical T-cell stage expressing CD4 or CD8, *WNK1* overexpression may reflect a lineage-specific dependency. Kochl et al³² demonstrated that thymocytes require *WNK1* and its effectors OXSR1, STK39, and SLC12A2 to upregulate MYC and to progress from double-negative 4 subsets into CD4⁺/CD8⁺ DP cells.

Figure 7 (continued) either with vehicle (DMSO) or WNK463 in Jurkat cells. Pearson linear regression model is shown. NT = nontargeting and #920 = shRNA directed against *WNK1*. (E) Heat map showing single-sample gene set enrichment analysis (ssGSEA) results enrichment MsigDB biological processes associated with *WNK1* inhibition after WNK463 treatment (2.5 μ M and 5 μ M) in Jurkat (sensitive) and DND-41 (resistant) cell lines. The heat map color scale indicates ssGSEA enrichment score that represents the degree to which the genes in a particular gene set are coordinately upregulated or downregulated within a sample. (F) Representative immunofluorescence images showing the effect on the mitotic spindle following *WNK1* loss in Jurkat-Cas9 cells 7 days after sgRNA selection. Cells were stained with anti-pericentrin (red) and anti- α -tubulin (green) antibodies, and nuclei with DAPI (blue). Scale bar, 10 μ m. NT = nontargeting, #1.1 = sgRNA directed against *WNK1*. (G) Histograms show the number of mitoses (black) and the number of polyploid nuclei (red) in Jurkat-Cas9 cells at 7 days after sgRNA selection. Statistical significance among groups ($***P \leq .001$) was determined by population statistics analysis using Fisher's exact test. NT = nontargeting, #1.1 = sgRNA directed against *WNK1*. (H) Representative fluorescent *in situ* hybridization (FISH) patterns reveal a 2.6-fold increase of polyploid cell nuclei in Jurkat-Cas9 cells at 7 days after sgRNA #1.1 selection using XL *DLEU/LAMP/12cen* metasytems probes. NT = nontargeting, #1.1 = sgRNA directed against *WNK1*. Original magnification $\times 100$. (I) Representative immunofluorescence images showing the effect on the mitotic spindle following *WNK1* inhibition in Jurkat cells 24 hours after WNK463 treatment. Cells were stained with anti-pericentrin (red) and anti- α -tubulin (green) antibodies, and nuclei with DAPI (blue). Scale bar, 10 μ m. (J) Histograms show the number of mitoses (black) and the number of polyploid nuclei (red) in Jurkat cells after WNK463 treatment. Statistical significance among groups ($*P \leq .05$; $**P \leq .01$) was determined using the Fisher exact test. (K) Representative FISH patterns reveal a 12-fold increase of polyploid cell nuclei in Jurkat cells after WNK463 treatment using *LSI BCR/ABL* (Vysis) and XL *1p32/1q21* (metasytems) probes. Original magnification $\times 100$.

Consistently, conditional genetic ablation studies have shown that WNK1-dependent water influx and volume regulation are required for CD4⁺ T-cell activation and T-cell-dependent antibody responses,⁶¹ further supporting a physiological requirement for WNK1 in T cells. Importantly, WNK1 activity is not necessarily inferred from abundance: we detect WNK1 in normal thymus but little/no phospho-WNK1, whereas phospho-WNK1 is more robust in T-ALL BM biopsies, indicating posttranslational activation of WNK1 signaling in leukemia.

Furthermore, WNK1 has been shown to control the homing of CD4⁺ and CD8⁺ T cells to the lymph nodes and spleen,⁴⁰ and part of these processes are mediated by CCR7, which regulates WNK1 activation, in turn controlling water homeostasis.⁶² CCR7 is an essential mediator for central nervous system invasion⁶³ in T-ALL as well as in multiple cancers.⁶⁴ Thus, one might speculate that WNK1 supports central nervous system infiltration in T-ALL, contributing to its malignant behavior.

Although WNK463 showed promise in treating cardiovascular conditions, its clinical development was halted due to cardiorenal toxicity observed in hypertensive rats at 10 mg/kg,³⁶ including ataxia and respiratory issues in mice.⁶⁵ In our study, no such adverse effects were seen; animals were sacrificed at planned end points without signs of WNK1-related toxicity. This may reflect differences in experimental models^{36,65} or a higher WNK1 dependency in T-ALL cells vs normal tissues, consistent with differential expression in BM and lymphoid samples. Nonetheless, subtle effects on electrolyte balance cannot be excluded and may have been mitigated by hydration. WNK463 remains a tool compound, highlighting the need for improved inhibitors.

Significant efforts have focused on optimizing WNK inhibitors due to their role in blood pressure regulation. WNK kinases are structurally distinct, with the catalytic lysine (Lys233 in WNK1) positioned outside the active site, unlike most kinases. This unique feature has enabled the development of WNK-specific ATP-competitive inhibitors and allosteric modulators. WNK463, for instance, is an orthosteric inhibitor that binds the ATP hinge region and accesses WNK1's back pocket via a narrow tunnel, shaped by the atypical positioning of Lys233 within the glycine-rich loop.⁶⁶ In contrast, WNK-IN-11 is a non-ATP-competitive inhibitor that binds an allosteric pocket near the ATP-binding site.³⁷ Of note, WNK-IN-11 has been originally reported to exhibit nearly 1000-fold selectivity for WNK1 over WNK4 and 57-fold selectivity for WNK1 over WNK2.³⁷ However, although WNK-IN-11 has been proposed as an alternative to WNK463, our studies did not confirm greater potency in T-ALL, suggesting that allosteric WNK1 inhibitors may require further optimization for this disease.³⁸

Small molecules targeting downstream elements of the WNK1 pathway have also been explored. For instance, the structurally related compounds closantel and STOCK1S-14279 inhibit SPAK kinase in an ATP-insensitive manner.⁶⁷ Rafoxanide has also been identified as a potent inhibitor of an allosteric pocket in the C-terminal (CCT) domains of SPAK/OSR1.^{68,69} Given that the CCT domain mediates the interaction between SPAK/OSR1 and WNK kinases through the RFxV[V/I] tetrapeptide sequence, which leads to SPAK/OSR1 activation, extensive

small-molecule screening has led to the identification of 2 compounds that disrupt this interaction.^{70,71}

Although none of the small molecules has been tested in T-ALL, WNK1 suppression has shown effects in solid tumors, where WNK1 is implicated in PI3K-AKT, transforming growth factor β , and NF- κ B signaling pathways, influencing cell migration and angiogenesis.^{72,73} For example, WNK463 has shown activity in breast cancer,⁷⁴ renal cancer,⁷⁵ and glioma.¹² Rafoxanide has been shown to inhibit growth, invasion, and migration in non-small cell lung cancer preclinical models⁷⁶ and has been linked to cell cycle regulation and modulation of the PTEN/PI3K/AKT pathway in skin cancer⁷⁷ and B-cell lymphoma.⁷⁸ However, the poor pharmacokinetic properties of these drugs pose challenges for their direct application in clinical trials without further optimization.³⁸

A key finding from our study is the correlation between WNK1 expression and the TAL-R patient cohort, where co-occurring mutations in *PTEN* often accompany del(6q), a chromosomal alteration that may signify a late-stage leukemogenesis event.⁷⁹ Because these genetic abnormalities are associated with poor prognosis, identifying WNK1 could enable personalized therapeutic strategies for patients carrying these alterations. However, no significant correlation was observed between P-WNK1 levels and sensitivity to WNK1 inhibition, suggesting that additional factors, such as PTEN status or broader signaling context, may be required for patient stratification. The precise mechanisms of WNK1 activation, its role across T-ALL subtypes, and the identification of predictive biomarkers remain to be fully defined.

In conclusion, the integration of genomic and proteomic dependency screens identifies WNK1 as a key driver of leukemogenesis in TAL-R T-ALL. Alongside previous evidence highlighting the role of atypical proteomes in carcinogenesis,⁸⁰ our findings further support the involvement of proteins regulating ion homeostasis⁸¹ in leukemia proliferation, paving the way for potential therapeutic interventions.

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Footnotes

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Original data are available from the corresponding author, Giovanni Roti (giovanni.roti@unipr.it), on request.

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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